

Employee Attrition Prediction – Project Summary

Executive Summary

Employee attrition is a major challenge for organizations because replacing experienced employees requires significant time, effort, and financial resources. The objective of this project was to develop a predictive system that helps Human Resources (HR) identify employees who may be at risk of leaving the company before they resign. Early identification enables HR teams to take proactive steps to improve employee satisfaction and retention.

The project analyzed employee information such as age, department, job role, monthly income, work-life balance, overtime, job satisfaction, years at the company, business travel, and other work-related factors. After cleaning and preparing the data, multiple prediction models were evaluated to determine the most suitable approach for identifying employees who are likely to leave. The selected model was then integrated into a user-friendly web application that allows HR personnel to enter employee details and instantly receive a prediction along with the probability of attrition.

The analysis showed that employee attrition is influenced by several factors rather than salary alone. Job role, overtime, business travel, job satisfaction, work-life balance, and career growth opportunities were among the most important factors affecting employee retention. Employees with lower job satisfaction, frequent overtime, and limited promotion opportunities showed a higher likelihood of leaving the organization.

The developed HR Decision Support System provides more than just a prediction. It also presents the estimated risk level and suggests practical recommendations that HR teams can use, such as reviewing workload, improving work-life balance, discussing career development, or considering compensation adjustments. These insights help HR departments make informed decisions and focus retention efforts on employees who are at greater risk.

Overall, this project demonstrates how data-driven decision making can support employee retention strategies. By identifying potential attrition in advance, organizations can reduce recruitment costs, improve employee engagement, and maintain a stable and productive workforce. The system serves as a practical HR analytics tool that can assist organizations in improving workforce planning and reducing voluntary employee turnover.